

F Report to Image-Level Filters



[Responsible Disclosure] Limitations of Marqo-NSFW on Politically Sensitive Content

Sun, May 17, 2026 at 5:17 PM

marqo.zip

Dear Marqo Team,

We are researchers in generative-AI safety. In a recent study, we introduce PC², a black-box jailbreaking framework that combines Identity-Preserving Descriptive Mapping (IPDM) with geopolitically distal multilingual prompts to bypass safety filters of commercial text-to-image (T2I) systems. Across GPT-4o, GPT-5, and GPT-5.1, PC² produces Politically Sensitive Contents (PSCs), photorealistic images of real public figures in fabricated political contexts such as heads of state holding extremist flags or performing ideologically charged gestures, with attack success rates up to 86%, while all original prompts are fully blocked. We are reaching out as part of our responsible-disclosure process.

To assess image-level defenses, we evaluated four open-source filters, including Marqo-NSFW (Marqo/nsfw-image-detection-384), on 207 PC²-generated PSCs and 207 benign political images. Marqo-NSFW achieved 5.30% TPR at FPR=0.1%, 7.25% at FPR=1%, and 14.50% at FPR=5%. This pattern, together with how the detections cluster on NSFW-adjacent texture, suggests that the gap is not one that threshold calibration can close, but a coverage gap rooted in the NSFW-only training objective. Inspecting the small fraction it flags, the detections appear to be driven by surface-level visual cues such as skin-tone distributions, distinctive chromatic patterns, or partial nudity, rather than by the political semantics of the scene (e.g., the combination of a recognizable public figure with an extremist symbol). In other words, the catches behave more like incidental noise on NSFW-adjacent texture than intentional recognition of PSCs.

We do not believe a single filter is expected to solve this on its own. In our paper we propose a layered defense pairing (i) a pre-filter we call relevant-language translation, which identifies the political entity in each adversarial prompt and converts the prompt into the language most relevant to that entity before moderation, with (ii) an image-level post-filter that understands political relations. Together, the two layers reduce PC²'s end-to-end attack success rate to roughly 11%. We would be glad to share our PSC-aware policy definition (covering political-figure impersonation and disinformation-via-visual-narrative as explicit categories) as a starting point for extending Marqo-NSFW's coverage to politically motivated misuse, and to discuss evaluation data that may help if you consider an expanded label space. For your reference, we have attached representative example images generated by PC² along with the corresponding adversarial prompts so that you can directly inspect the failure modes described above. Please let us know if you would like to coordinate further.

Sincerely,

Marqo
(a)



[Responsible Disclosure] Limitations of Falconsai NSFW Detector on Politically Sensitive Content

Sun, May 17, 2026 at 5:13 PM

falconsai.zip

Dear Falconsai Team,

We are researchers in Generative AI security. In a recent study, we introduce PC², a black-box jailbreaking framework that combines Identity-Preserving Descriptive Mapping (IPDM) with geopolitically distal multilingual prompts to bypass safety filters of commercial text-to-image (T2I) systems. Across GPT-4o, GPT-5, and GPT-5.1, PC² produces what we call Politically Sensitive Contents (PSCs), photorealistic images of real public figures in fabricated political contexts such as heads of state holding extremist flags or performing ideologically charged gestures, with attack success rates up to 86%, while all original prompts are fully blocked. Such content poses concrete risks in active conflicts and election cycles, and we are reaching out as part of our responsible-disclosure process.

To assess image-level defenses, we evaluated four open-source filters, including Falconsai (Falconsai/nsfw_image_detection), on 207 PC²-generated PSCs and 207 benign political images. Among the filters tested, Falconsai showed the most limited recognition of PSCs: 0.00% TPR at FPR=0.1%, 0.00% at FPR=1%, and 1.40% at FPR=5%. This pattern suggests not a threshold-calibration issue but a coverage gap rooted in the training objective. Because the model is optimized solely for traditional NSFW categories (nudity and explicit content), it has no representation of political identity, geopolitical symbolism, or relational political harm, and the few PSCs it does flag appear to be incidental rather than semantically grounded.

We do not believe a single filter is expected to solve this on its own. In our paper we propose a layered defense pairing (i) a pre-filter we call relevant-language translation, which identifies the political entity in each adversarial prompt and converts the prompt into the language most relevant to that entity before moderation, with (ii) an image-level post-filter that understands political relations. Together, the two layers reduce PC²'s end-to-end attack success rate to roughly 11%. We would be glad to share our PSC-aware policy definition (covering political-figure impersonation and disinformation-via-visual-narrative as explicit categories) as a starting point for extending Falconsai's coverage beyond traditional NSFW, and to discuss data or evaluation protocols that may help if you choose to expand the training distribution. For your reference, we have attached representative example images generated by PC² along with the corresponding adversarial prompts so that you can directly inspect the failure modes described above. Please let us know if you would like to coordinate further.

Sincerely,

Falconsai
(a)



[Responsible Disclosure] Limitations of the SD Safety Checker on Politically Sensitive Content

Sun, May 17, 2026 at 5:21 PM

SDsafetychecker.zip

Dear Stable Diffusion Safety Checker / CompVis Team,

We are researchers in generative-AI safety. In a recent study, we introduce PC², a black-box jailbreaking framework that combines Identity-Preserving Descriptive Mapping (IPDM) with geopolitically distal multilingual prompts to bypass safety filters of commercial text-to-image (T2I) systems. Across GPT-4o, GPT-5, and GPT-5.1, PC² produces Politically Sensitive Contents (PSCs), photorealistic images of real public figures in fabricated political contexts such as heads of state holding extremist flags or performing ideologically charged gestures, with attack success rates up to 86%, while all original prompts are fully blocked. We are reaching out as part of our responsible-disclosure process.

To assess image-level defenses, we evaluated four open-source filters, including the Stable Diffusion safety checker (CompVis/stable-diffusion-safety-checker), on 207 PC²-generated PSCs and 207 benign political images. The SD safety checker performed best among the classifier-based filters we tested, but still missed the large majority of PSCs: 5.31% TPR at FPR=0.1%, 15.46% at FPR=1%, and 33.30% at FPR=5%. Examining its detections, the model appears to flag PSCs only when they incidentally trigger NSFW concept embeddings, for example images with high skin exposure, blood-like chromatic patterns, or weapon silhouettes, rather than recognizing the politically sensitive relation between the depicted figure and the surrounding symbols. The detections are therefore better characterized as coincidental NSFW-adjacent matches than as semantically grounded PSC recognition.

We do not believe a single filter is expected to solve this on its own. In our paper we propose a layered defense pairing (i) a pre-filter we call relevant-language translation, which identifies the political entity in each adversarial prompt and converts the prompt into the language most relevant to that entity before moderation, with (ii) an image-level post-filter that understands political relations. Together, the two layers reduce PC²'s end-to-end attack success rate to roughly 11%. We would be glad to share our PSC-aware policy definition (covering political-figure impersonation and disinformation-via-visual-narrative as explicit categories) and to discuss how additional concept embeddings or auxiliary classifiers might extend the safety checker beyond its current NSFW-only scope. For your reference, we have attached representative example images generated by PC² along with the corresponding adversarial prompts so that you can directly inspect the failure modes described above. Please let us know if you would like to coordinate further.

Sincerely,

SD Safety Checker
(b)



[Responsible Disclosure] Limitations of the LlavaGuard on Politically Sensitive Content

Sun, May 17, 2026 at 5:30 PM

llavaguard.zip

Dear LlavaGuard Team,

We are researchers in generative-AI safety. In a recent study, we introduce PC², a black-box jailbreaking framework that combines Identity-Preserving Descriptive Mapping (IPDM) with geopolitically distal multilingual prompts to bypass safety filters of commercial text-to-image (T2I) systems. Across GPT-4o, GPT-5, and GPT-5.1, PC² produces Politically Sensitive Contents (PSCs), photorealistic images of real public figures in fabricated political contexts such as heads of state holding extremist flags or performing ideologically charged gestures, with attack success rates up to 86%, while all original prompts are fully blocked. We are reaching out as part of our responsible-disclosure process.

To assess image-level defenses, we evaluated four open-source filters, including LlavaGuard (AIML-TUDA/LlavaGuard-v1.2-7B-OV-Hf), on 207 PC²-generated PSCs and 207 benign political images. Under its default policy, LlavaGuard reached 15.94% TPR at 0% FPR, the strongest result among the filters tested, reflecting the advantage of VLM-based reasoning, yet still well below what is needed to mitigate PSCs in practice. Examining its outputs, the few PSCs it does catch are classified almost exclusively under T1 (Hate, Humiliation), T7 (Radicalization), T9 (Disinformation), or T10 (Hate Symbols), with rationales grounded in overtly hateful visual elements (e.g., a tank, a Nazi flag) rather than the relational political meaning that actually constitutes the harm. For instance, a Nazi salute performed by a head of state before a public crowd is described as "a person in a suit waving a hand" and labeled Safe. The default policy also tends to default to Safe whenever the figure wears formal attire or performs a ceremonial-looking gesture.

Because LlavaGuard accepts a policy at inference time, we explored whether the gap can be closed without retraining. We augmented the default policy with an additional category, T11: Politically Sensitive Content, that explicitly defines political-figure impersonation with extremist symbols, ideologically charged salutes, and fabricated attributions of contested claims as unsafe (full prompt available on request). The resulting variant, LlavaGuard-PSC, reaches 47.83% TPR at 0.48% FPR, a +31.89pp gain over the default policy. Category-wise analysis shows that 84 of the new detections fall under the added T11 category, recovering cases the default policy missed by reasoning in isolation about each visual element. We pair this post-filter with a text-level pre-filter (relevant-language translation), which identifies the political entity in each adversarial prompt and converts the prompt into the language most relevant to that entity before moderation, and the layered combination reduces PC²'s end-to-end attack success rate to roughly 11%. Notably, when politically charged phrases appear inside the image in a non-English language (e.g., a placard or document rendered in the target country's script), LlavaGuard-PSC frequently fails to read or interpret them and defaults to Safe, allowing the PSC to pass even when the policy itself is satisfied in principle. Residual failures expose three remaining capability gaps: identity grounding (PSC detection presupposes recognizing the figure), cross-lingual grounding (non-English in-image text defaults to Safe), and counterfactual reasoning (contested territorial slogans read as factual), which may be useful signals for future LlavaGuard versions.

We would be very glad to share the full PSC-aware policy and our evaluation set, and to coordinate on incorporating PSC coverage into upstream LlavaGuard. For your reference, we have attached representative example images generated by PC² along with the corresponding adversarial prompts so that you can directly inspect the failure modes described above. Please let us know if you would like to discuss further.

Sincerely,

LlavaGuard
(b)

Figure 16: Report emails for image-level filters.